
6a	From Fig. 6.2, magnetic flux density = 50 mT Magnetic flux linkage, Φ = NBA $= 180 \times 50 \times 10^{-3} \times \pi (5.3 \times 10^{-3} / 2)^2$ $= 1.9856 \times 10^{-4} = 2.0 \times 10^{-4}$ Wb Note: To get the answer mark, the calculated value must be written down before showing the answer to be that in the question.	M1 M1 A1

6b(i)	change in magnetic flux density ΔB = 5 – 50 = – 45 mT (from graph, accept positive value) average induced e.m.f. = $\Delta\Phi / \Delta t = \Delta NBA / \Delta t = NA\ \Delta B / \Delta t$ $= 180 \times \pi\ (5.3\times10^{-3}/2)^2 \times(-45\times10^{-3}) / 0.30$ $= -5.957\times10^{-4} = -6.0\times10^{-4}$ V Note: accept if answer is given as a positive value	C1 A1
6b(ii)	The change in magnetic flux linkage / magnetic flux / magnetic flux density decreases as distance increases. **OR** At constant speed, the rate of change of magnetic flux linkage / induced e.m.f. decreases as x increases. Hence to keep the e.m.f. constant, the speed must increase.	M1 A1
6c	As the coil moves, there is changing / decreasing flux linkage in the coil, leading to an induced e.m.f. Induced e.m.f. results in induced current which creates a magnetic field / flux in the coil that opposes the motion of the coil. As work is done / energy is needed to move the coil away from the magnet, (induced) current gives the heating effect (in the resistor) which comes from the work done. **OR** (last two points can be replaced by the following) Work done in moving the coil is converted to electrical energy. When induced current flows through resistor, electrical energy is converted to heat.	B1 B1 B1 B1 B1 B1
7a	Electric field strength is defined as the electric force per unit positive charge placed at	B2